Comparing Time Series Forecasting Methods for the U.S. Gasoline-Diesel Price Spread

1. Abstract

Fuel prices significantly affect transportation costs, consumer spending, and freight industries in the United States. Gasoline is primarily consumed by passenger vehicles while diesel powers commercial freight transportation, making the price spread between the two fuels an important economic indicator. Understanding and forecasting this spread can provide valuable insights for businesses, policymakers, and consumers who are sensitive to fuel price fluctuations.

This project investigates the weekly gasoline–diesel price spread in the United States, defined as the difference between the U.S. All Grades All Formulations retail gasoline price and the diesel retail price, measured in dollars per gallon. The dataset spans January 1995 to December 2021 with 1,361 weekly observations obtained from the U.S. Energy Information Administration (EIA). The goal is to determine which time series forecasting framework, exponential smoothing, ARIMA, or ARIMAX, provides the most accurate short-term forecasts of the spread, evaluated using rolling 1-step-ahead holdout RMSE over a 104-week holdout period.

The spread exhibits clear non-stationarity, with a structural shift around 2008 after which diesel prices have consistently exceeded gasoline prices. The ACF and PACF of the differenced spread confirm that first differencing is sufficient to achieve stationarity.

Two baseline rules were evaluated first. The training mean achieved an **RMSE of 0.246** and the persistence rule achieved an **RMSE of 0.029**. All time series models improved upon both baselines. Among exponential smoothing models, Holt linear smoothing performed best with an **RMSE of 0.025**, outperforming seasonal Holt-Winters variants with periods of 26 and 52 weeks. For ARIMA, a systematic search over candidate models identified ARIMA(4,1,3) as the best performer with an RMSE of 0.023, improving upon both baselines and exponential smoothing. An ARIMAX model incorporating lagged crude oil and heating oil prices as external regressors matched but did not improve upon the best ARIMA model, also achieving an RMSE of 0.023, suggesting that the spread's own history is sufficient for short-term forecasting.

Overall, ARIMA(4,1,3) is recommended as the preferred forecasting model for the gasoline–diesel price spread, offering the best balance of accuracy and efficiency.

2. Exploratory Data Analysis

The dataset is obtained from the U.S. Energy Information Administration (EIA) and is publicly available via Kaggle (https://www.kaggle.com/datasets/mruanova/us-gasoline-and-diesel-retail-prices-19952021). It contains weekly retail fuel prices across multiple gasoline grades and formulations, as well as diesel prices, spanning from January 1995 to December 2021, yielding 1,361 weekly observations. All prices are measured in dollars per gallon. No missing values or gaps in the weekly frequency were found.
The response variable is the weekly price spread, defined as:

Spread_t = Gasoline_t − Diesel_t

where Gasoline_t refers to the weekly U.S. All Grades All Formulations retail gasoline price. This spread captures the relative pricing dynamics between the two fuels and serves as the target variable for forecasting.
Initial exploration of gasoline prices across different grades and formulations shows that all series follow highly similar trends over time, indicating strong correlation among them (see Appendix). Therefore, the all-grades, all-formulations gasoline series is used as a representative measure for the analysis.

To focus on the relationship between gasoline and diesel, the spread series is constructed and plotted over time. The plot shows noticeable variation over the sample period, with a structural shift around the late 2000s where gasoline prices become consistently lower than diesel. The series also exhibits short-term fluctuations, suggesting potential temporal dependence.

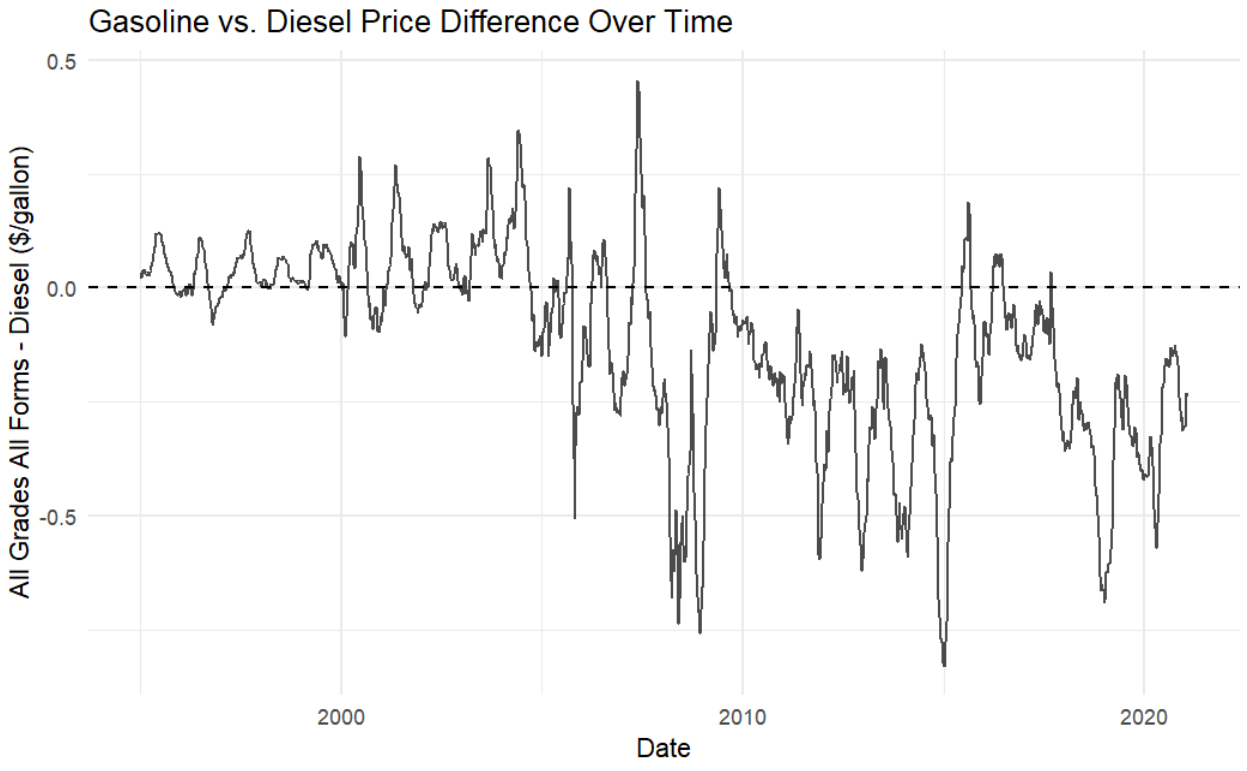


Figure 1: Weekly gasoline–diesel price spread ($/gallon), January 1995 – December 2021

Table 1: Summary statistics for key variables, 1995–2021.

Variable	Mean	SD	Min	Max
Gasoline ($/gal)	2.286	0.859	0.949	4.165
Diesel ($/gal)	2.405	0.999	0.953	4.764
Spread ($/gal)	−0.119	0.215	−0.829	0.455
Log Ratio	−0.032	0.084	−0.358	0.184

Summary statistics provide additional insight into the scale and variability of the data. Gasoline and diesel prices have similar magnitudes, while the spread has a relatively small mean and lower variance, indicating that it fluctuates within a narrower range. The log ratio between gasoline and diesel prices shows similar behavior, suggesting that the relative difference between the two series remains modest over time.

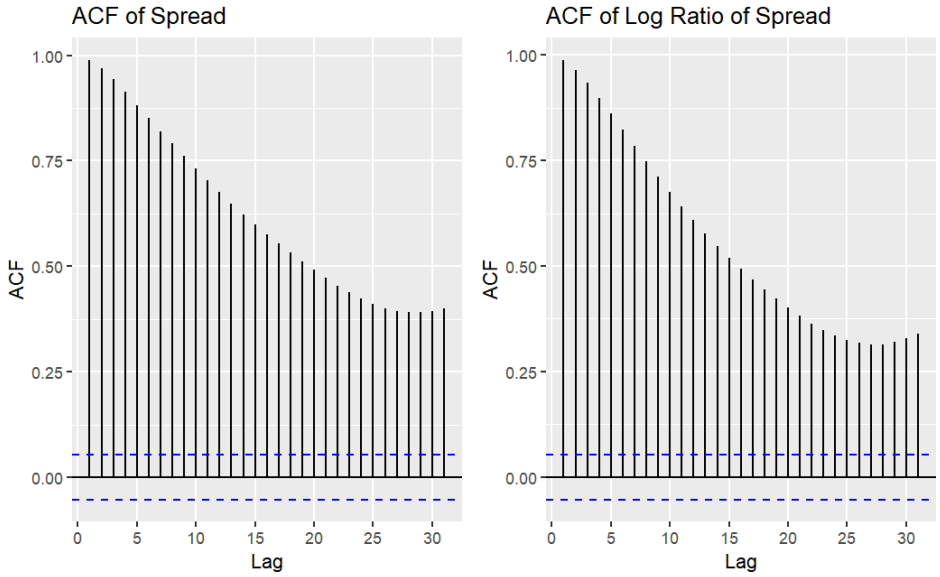


Figure 2: ACF of the price spread (left) and ACF of the log of Spread (right)

To assess whether a transformation improves stationarity, the autocorrelation functions (ACF) of both the spread and the logarithm of the spread are compared. Both ACF plots exhibit a slow decay across multiple lags, indicating strong persistence and non-stationarity in each series.

The ACF of the differenced series shows a strong autocorrelation at lag 1 followed by a rapid decay, indicating short-term dependence. The partial autocorrelation function (PACF) exhibits a dominant spike at lag 1, suggesting that an autoregressive structure may be appropriate for modeling. A log-differenced version of the series was also examined (see Appendix), but it did not provide any improvement in the autocorrelation structure. Therefore, the first-differenced spread is retained for further analysis.

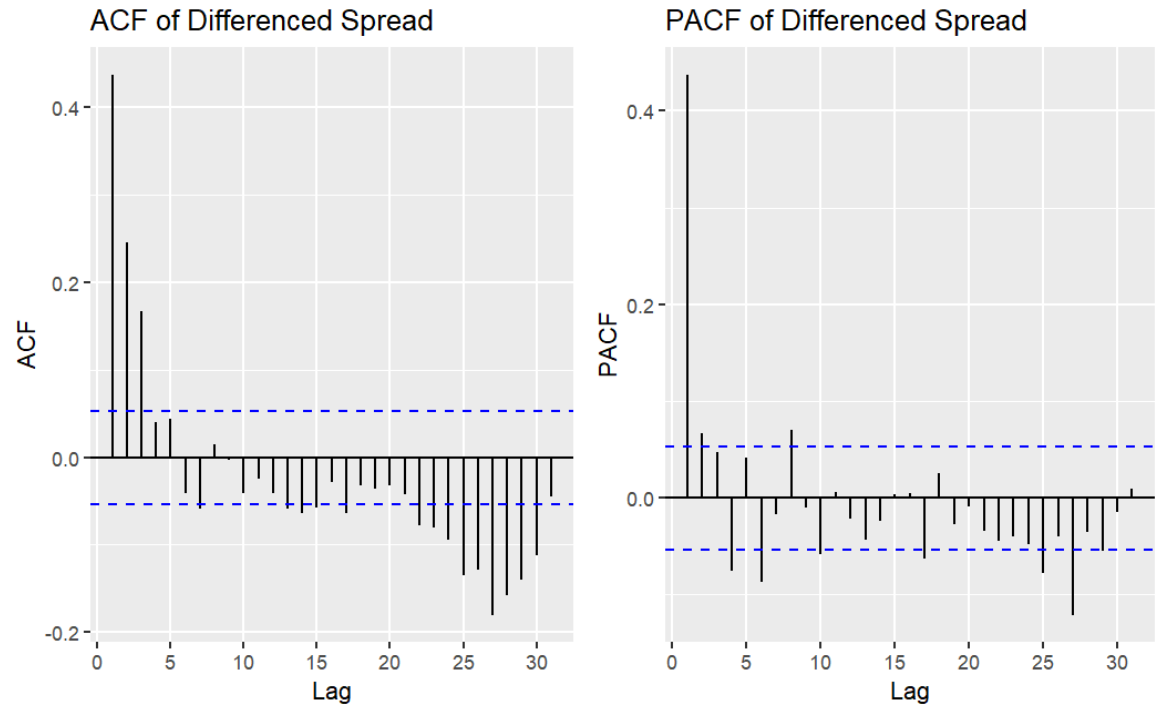


Figure 3: ACF of the differenced price spread (left) and PACF of the differenced price Spread (right)

To investigate potential lead–lag relationships between gasoline and diesel prices, the cross-correlation function (CCF) of the differenced series is examined. The CCF exhibits a strong peak at lag 0, indicating that changes in gasoline and diesel prices occur contemporaneously. While smaller correlations are present at nearby lags, there is no dominant peak away from zero, suggesting that neither series consistently leads or lags the other. This implies that lagged values of diesel (or gasoline) are unlikely to provide substantial additional predictive power in a forecasting model.

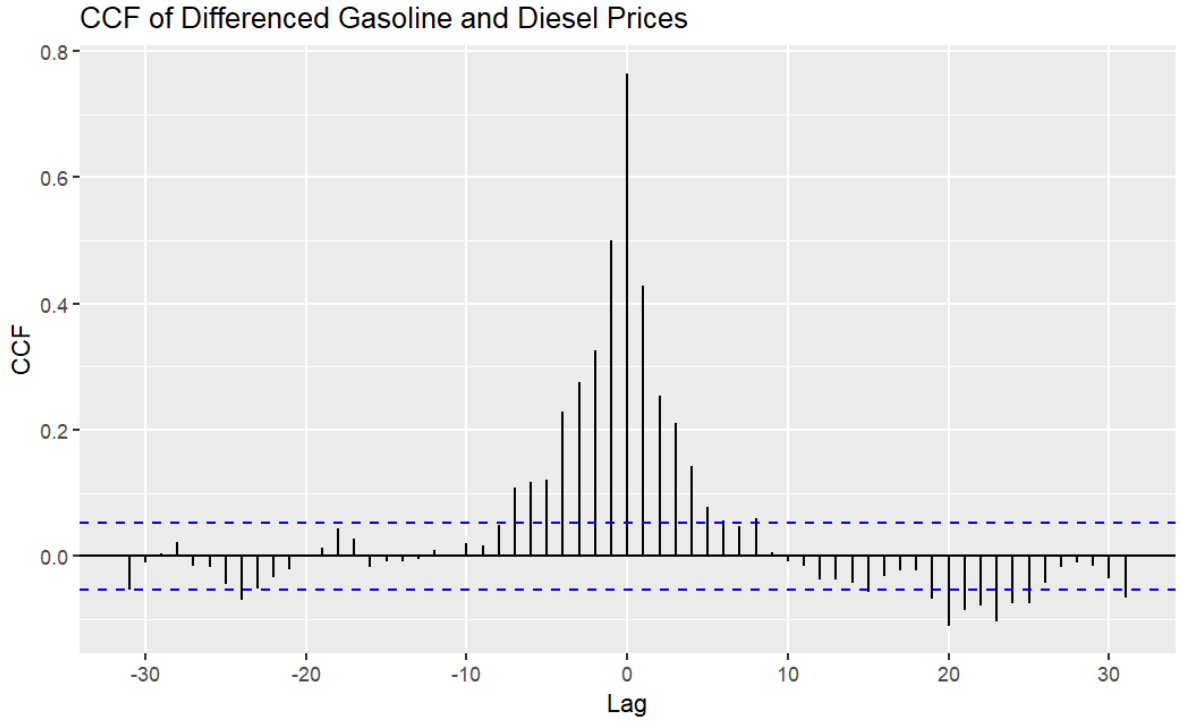


Figure 4: CCF of Differenced Gasoline and Diesel

3. Baseline Models

To evaluate the performance of more advanced forecasting methods, two simple baseline models are first considered: the persistence model and the IID mean model.
The persistence model uses the most recent observed value as the forecast:

$$\hat{y}_{t+1|t} = y_t$$

The IID model assumes that observations are independent over time and forecasts future values using the mean of the training data:

$$\hat{y}_{t+1|t} = \bar{y}_{train}$$

Both rules are evaluated using rolling 1-step-ahead forecasts over the 104-week holdout set. Results are summarized below:

Table 2: Baseline Model Performance (RMSE)

Model	RMSE
Persistence	0.0291
IID (Mean)	0.246

The persistence model substantially outperforms the training mean, reflecting the non-stationary nature of the spread. Because the level of the series evolves over time, the historical mean is a poor predictor. The strong performance of persistence highlights the importance of recent observations and provides a benchmark of RMSE of 0.0291 against which more advanced models can be evaluated.

4. Exponential Smoothing

Exponential smoothing methods are considered as a class of models that capture level and trend dynamics in time series data.Three exponential smoothing models were fitted to the training set and evaluated using rolling 1-step-ahead forecasts over the **104-week holdout period**.
Simple exponential smoothing (SES) was not considered as the EDA confirmed the presence of a trend in the spread series, making SES inappropriate. So, **Holt linear exponential smoothing** was applied, which accounts for both level and trend. In addition, seasonal Holt–Winters models with periods of 26 and 52 weeks are also evaluated to assess whether seasonal patterns improve forecasting.

Table 3: Exponential Smoothing Performance (RMSE)

Model	***RMSE***
Holt Linear	*0.0253*
Holt-Winters (26)	*0.0305*
Holt-Winters (52)	*0.0309*

The Holt linear model outperforms both seasonal variants and achieves the lowest RMSE among the smoothing methods. In contrast, the seasonal Holt–Winters models perform worse, indicating that seasonal components do not contribute meaningfully to forecasting accuracy. The best exponential smoothing model is **Holt Linear** with a holdout **RMSE of 0.0253**, which improves upon the persistence baseline of **0.029**.

5. ARIMA Models

The ACF of the spread series shows slow decay across all lags, confirming non-stationarity. First differencing produces a stationary series, as evidenced by the ACF of the differenced spread dropping quickly to zero. This justifies fitting ARIMA models with d = 1 and of the form ARIMA(p,1,q).

As a starting point, `auto.arima()` was applied to the training set, suggesting ARIMA(3,1,2) based on AIC. Since the model with the lowest AIC does not necessarily yield the best out-of-sample forecast, a systematic search was conducted over models with perturbed p and q values around the auto.arima suggestion, such as ARIMA(1,1,0), ARIMA(2,1,1), and ARIMA(4,1,3). Each model is assessed using rolling one-step-ahead forecasts on the holdout set.

Table 4: ARIMA candidate models ranked by holdout RMSE

	Model	Holdout_RMSE
1	ARIMA(4,1,3)	0.0232
2	ARIMA(2,1,1)	0.0238
3	ARIMA(2,1,3)	0.0238
4	ARIMA(3,1,3)	0.0238
5	ARIMA(4,1,1)	0.0238
6	ARIMA(4,1,2)	0.0238

Among the models considered, ARIMA(4,1,3) achieves the lowest holdout RMSE of 0.0232. This model outperforms both the persistence (0.0291) and exponential smoothing (0.0253) models, indicating that incorporating both autoregressive and moving average components improves forecasting accuracy.

Seasonal ARIMA models with periods of 26 and 52 weeks are also evaluated. However, adding seasonal components did not improve upon the best non-seasonal model. The top seasonal candidates ARIMA(4,1,3)(1,0,0)[26] and ARIMA(4,1,3)(1,0,0)[52] both achieved the same RMSE of 0.0232. The best ARIMA model is therefore **ARIMA(4,1,3)** with a holdout **RMSE of 0.0232**.

Residual diagnostics for the best model confirm its adequacy. The ACF of residuals (see Appendix) shows no significant autocorrelation remaining and the Ljung-Box test yields a p-value of 0.365, failing to reject the null hypothesis of white noise residuals.

6. ARIMAX Models

To investigate whether external variables can be used to improve forecasting performance, we considered ARIMAX models incorporating crude oil and heating oil prices as explanatory variables. These variables were selected due their close relation to fuel markets and may influence the gasoline-diesel price spread. Crude oil prices were obtained from the Federal Reserve Bank of St. Louis and heating oil prices were obtained from the U.S. Energy Information Administration.

The CCF plots of the differenced gas-diesel spread with each of the differenced crude oil prices and the differenced heating oil prices show the largest spike at lag -1, suggesting that a lag of one week would be the most appropriate to use for forecasting the gas-diesel spread.

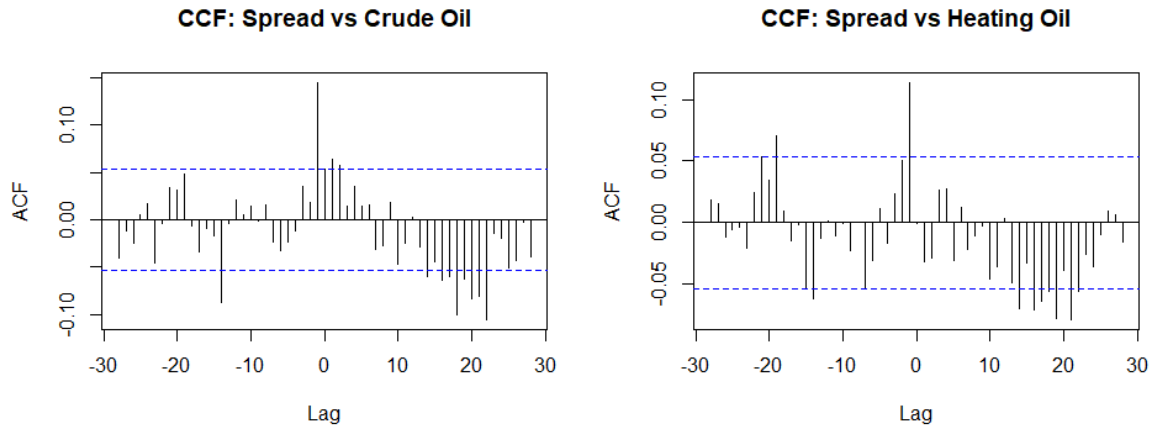


Figure 5 : Cross-correlation functions between the differenced spread and differenced crude

We then run linear regression models using the lagged crude oil prices, lagged heating oil prices, and both, as the three candidate regressors. The residuals of each linear model are then run through auto.arima separately. For the linear model using lagged heating oil as the explanatory variable, ARIMA(3,1,2) on the residuals is suggested; the same is suggested of the residuals of the linear model using crude oil price as the explanatory variable. For the linear model using both lagged heating and crude oil as predictors, ARIMA(2,0,2) on the residuals is residuals.
Based on these models suggested by auto.arima, we then run a search using perturbed ARIMA models on the residuals. Our search found that the best ARIMAX model used both the lagged crude and heating oil prices as explanatory variables, with ARIMA(2,0,1) on the residuals. This model achieved a holdout RMSE of 0.0232, which is tied in performance to the ARIMA(4,1,3) model mentioned above.

7. Forecast Comparison

All models were evaluated using rolling 1-step-ahead forecasts over the 104-week holdout set (January 2020 – December 2021), with model coefficients fixed at their training set estimates. Performance is measured by holdout Root Mean Squared Error (RMSE).

Table 5: Holdout RMSE for all forecasting models.

Model	Holdout RMSE
Training Mean	0.2460
Persistence	0.0291
Holt Linear	0.0253
Holt-Winters Additive Seasonal (period 52)	0.0309
Holt-Winters Additive Seasonal (period 26)	0.0305
ARIMA(4,1,3)	0.0232
ARIMAX: crude_lag1 + heat_lag1, ARIMA(2,0,1)	0.0232

All time series models substantially outperform the training mean baseline. The persistence rule achieves a competitive RMSE of 0.029, which all three model classes (exponential smoothing, ARIMA, ARIMAX) successfully improve upon.

The best performing models are ARIMA(4,1,3) and the ARIMAX model with crude oil and heating oil lags, both achieving a holdout RMSE of 0.0232. The Holt linear model performs reasonably well at 0.0253 but is outperformed by both ARIMA and ARIMAX. The seasonal exponential smoothing models perform worse than Holt linear, confirming that explicitly modeling seasonality does not benefit forecast accuracy for this series.

The ARIMAX model does not improve upon the best ARIMA model, suggesting that lagged crude oil and heating oil prices do not provide additional predictive power beyond what is already captured by the autoregressive structure of the spread series itself. Given the equivalent forecast accuracy, the simpler ARIMA(4,1,3) model is preferred for its parsimony.

8. Conclusion

This study analyzed the weekly gasoline–diesel price spread using several time series models. The exploratory analysis indicated non-stationarity, leading to the use of first differencing.

Among the models considered, ARIMA(4,1,3) achieved the lowest holdout RMSE and provided the best forecasting performance. Exponential smoothing improved upon simple baselines but was less accurate than ARIMA. ARIMAX models incorporating crude oil and heating oil did not produce meaningful improvements.

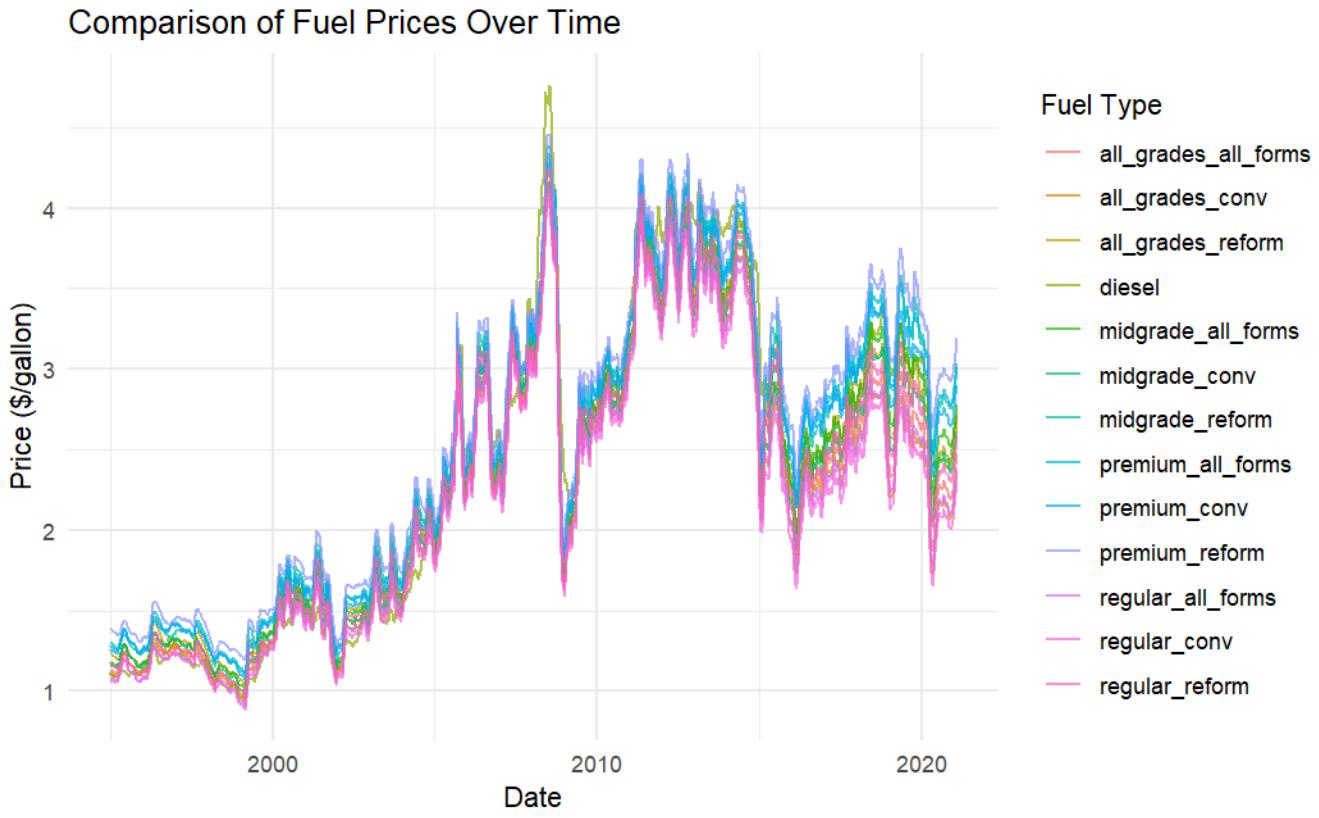
Overall, the results suggest that the spread is primarily driven by its own past values and ARIMA models are well-suited for capturing its dynamics.

Appendix A: Additional Exploratory Data Analysis

This appendix contains supplementary exploratory analysis that supports, but is not essential to, the main modeling decisions presented in the report.

A.1 Gasoline Price Series by Grade and Formulation

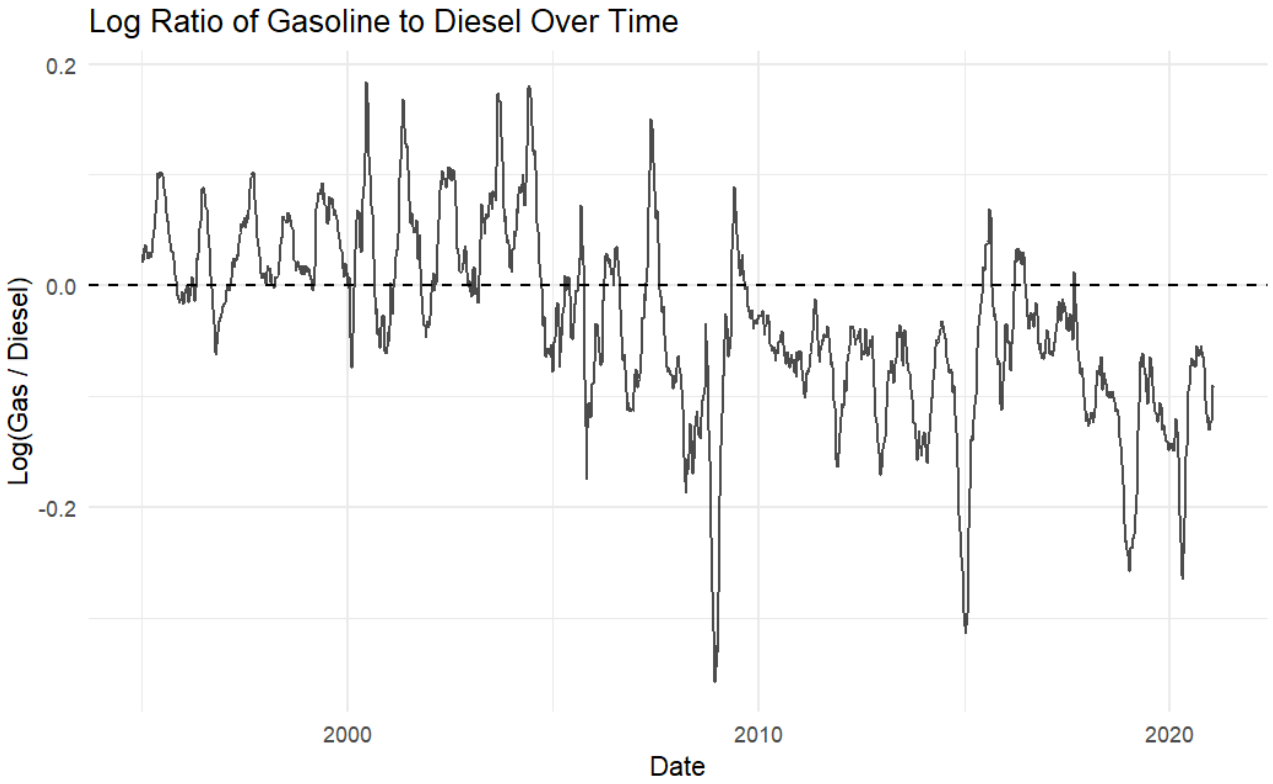
Plots comparing gasoline prices across different grades and formulations are provided to demonstrate that all gasoline series follow highly similar trends over time. This supports the use of the all-grades, all-formulations gasoline series as a representative measure in the main analysis.

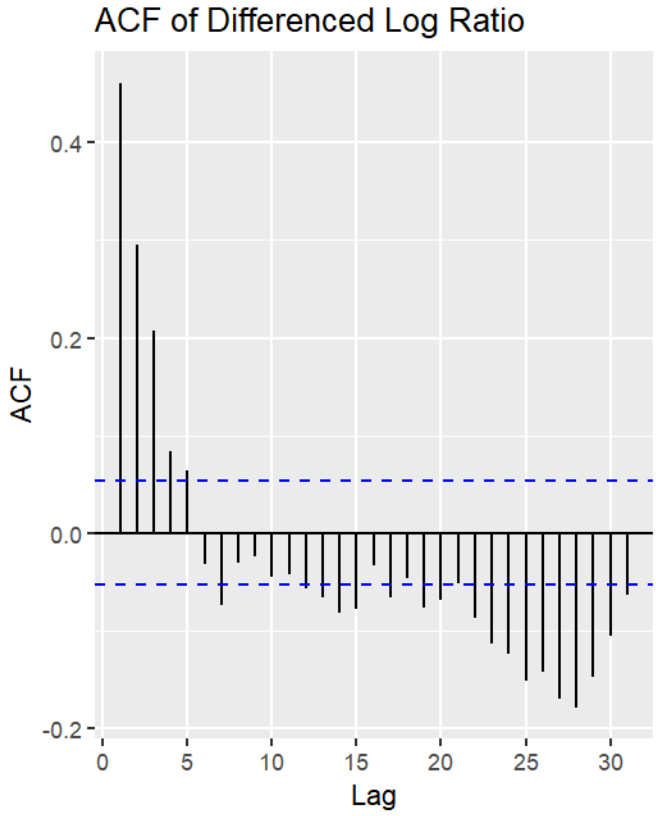


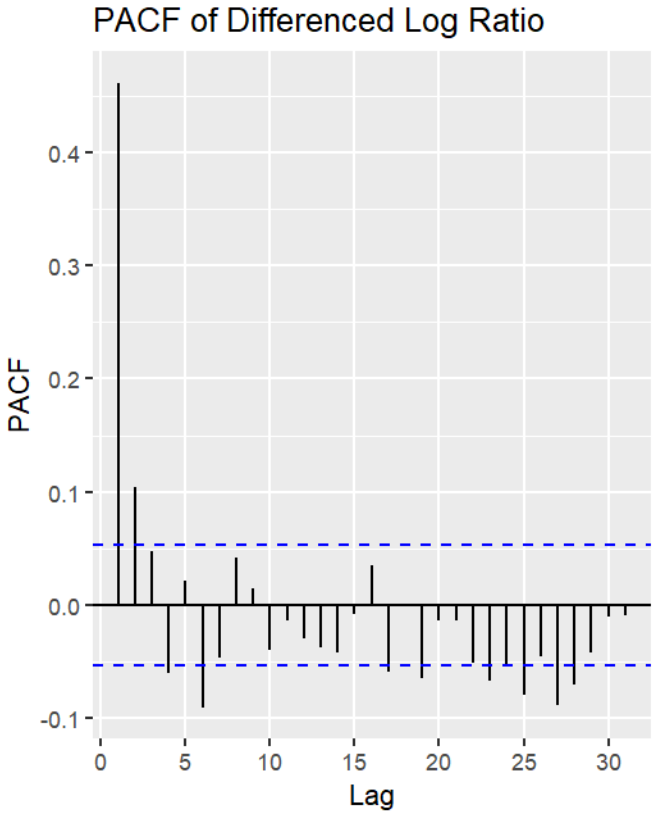
A.2 Log Ratio and Log-Differenced Series

The logarithm of the gasoline-to-diesel price ratio and its first difference were explored as potential transformations to stabilize variance and improve stationarity.

The autocorrelation structure of both the log-transformed and log-differenced series was found to be very similar to that of the original spread and differenced spread. As a result, these transformations did not provide additional modeling benefits and were not used in the final analysis.


Log Ratio of Gasoline to Diesel Over Time
0.2
0.0
-0.2
Log(Gas / Diesel)
2000
2010
2020
Date


ACF of Differenced Log Ratio
0.4
0.2
0.0
-0.2
ACF
0
5
10
15
20
25
30
Lag

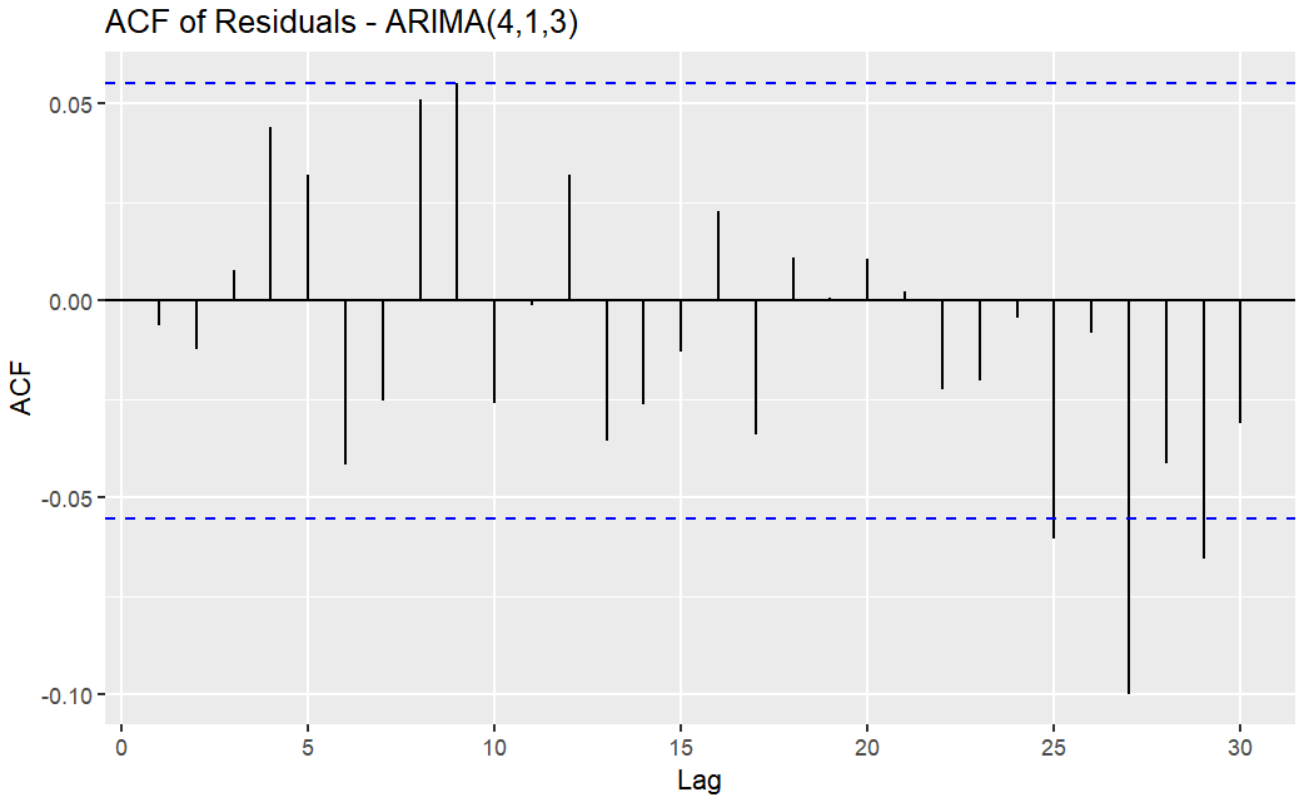

PACF of Differenced Log Ratio
0.4
0.3
0.2
0.1
0.0
-0.1
PACF
0
5
10
15
20
25
30
Lag

Appendix B: Residual Diagnostics

B.1 ARIMA Residual Analysis

Residual diagnostics for the selected ARIMA(4,1,3) model are provided to assess model adequacy.

The autocorrelation function (ACF) of the residuals shows no significant spikes outside the confidence bounds, indicating that the residuals are approximately uncorrelated. This supports the conclusion that the model has successfully captured the temporal dependence structure.



The Ljung–Box test further confirms this, with a p-value of 0.365, failing to reject the null hypothesis of white noise residuals

```
        Box-Ljung test

data:  best_arima$residuals
X-squared = 21.554, df = 20, p-value = 0.3652
```

Appendix C: Additional ARIMA Model Results

A full list of candidate ARIMA models evaluated during the model selection process is provided below. These models were assessed using holdout RMSE, though only the top-performing models are discussed in the main report.

	Model	Holdout_RMSE
1	ARIMA(4,1,3)	0.0232
2	ARIMA(2,1,1)	0.0238
3	ARIMA(2,1,3)	0.0238
4	ARIMA(3,1,3)	0.0238
5	ARIMA(4,1,1)	0.0238
6	ARIMA(4,1,2)	0.0238
7	ARIMA(1,1,1)	0.0238
8	ARIMA(3,1,1)	0.0239
9	ARIMA(1,1,2)	0.0239
10	ARIMA(3,2,2)	0.0239
11	ARIMA(2,1,2)	0.0240
12	ARIMA(3,1,2)	0.0240
13	ARIMA(3,2,3)	0.0240
14	ARIMA(1,1,0)	0.0242
15	ARIMA(2,2,2)	0.0243

Seasonal ARIMA candidates with holdout RMSE

	Model	Holdout_RMSE
1	ARIMA(4,1,3)(1,0,0)[26]	0.0232
2	ARIMA(4,1,3)(1,0,0)[52]	0.0232
3	ARIMA(4,1,3)(1,0,1)[26]	0.0233
4	ARIMA(4,1,3)(1,0,1)[52]	0.0236
5	ARIMA(3,1,2)(1,0,0)[52]	0.0240
6	ARIMA(1,1,0)(1,0,0)[26]	0.0242
7	ARIMA(1,1,0)(1,0,0)[52]	0.0242
8	ARIMA(3,1,2)(1,0,1)[52]	0.0243
9	ARIMA(0,1,1)(0,0,1)[26]	0.0259
10	ARIMA(0,1,1)(0,0,1)[52]	0.0260
11	ARIMA(3,0,2)(1,1,0)[26]	0.0283
12	ARIMA(2,0,1)(1,1,0)[26]	0.0285
13	ARIMA(4,0,3)(1,1,0)[26]	0.0290
14	ARIMA(2,0,1)(1,1,0)[52]	0.0296
15	ARIMA(4,0,3)(1,1,0)[52]	0.0296
16	ARIMA(3,0,2)(1,1,0)[52]	0.0297
17	ARIMA(1,0,1)(1,1,0)[52]	0.0317

Appendix D: ARIMAX Supporting Analysis

D.1 Full table of all ARIMAX candidates with holdout RMSE

	Exogenous_Variable	Holdout_RMSE
1	crudeheat_lag1 with ARIMA(2,0,1) on residuals	0.0232
2	crude_lag1 with ARIMA(4,1,3) on residuals	0.0233
3	crude_lag1 with ARIMA(3,0,3) on residuals	0.0233
4	crudeheat_lag1 with ARIMA(4,1,3) on residuals	0.0233
5	crudeheat_lag1 with ARIMA(3,0,3) on residuals	0.0233
6	crude_lag1 with ARIMA(2,0,1) on residuals	0.0234
7	crudeheat_lag1 with ARIMA(2,0,2) on residuals	0.0234
8	crudeheat_lag1 with ARIMA(3,0,2) on residuals	0.0234
9	crudeheat_lag1 with ARIMA(2,0,3) on residuals	0.0234
10	heat_lag1 with ARIMA(2,0,1) on residuals	0.0235
11	crude_lag1 with ARIMA(2,0,2) on residuals	0.0235
12	crude_lag1 with ARIMA(3,0,2) on residuals	0.0235
13	crude_lag1 with ARIMA(2,0,3) on residuals	0.0235
14	heat_lag1 with ARIMA(4,1,3) on residuals	0.0236
15	heat_lag1 with ARIMA(3,0,3) on residuals	0.0236
16	heat_lag1 with ARIMA(2,0,2) on residuals	0.0237
17	heat_lag1 with ARIMA(3,0,2) on residuals	0.0237
18	heat_lag1 with ARIMA(2,0,3) on residuals	0.0237
19	crude_lag1 with ARIMA(3,1,3) on residuals	0.0237
20	crudeheat_lag1 with ARIMA(3,1,3) on residuals	0.0237
21	crude_lag1 with ARIMA(3,1,1) on residuals	0.0238
22	crudeheat_lag1 with ARIMA(3,1,1) on residuals	0.0238

```
23     crude_lag1 with ARIMA(3,1,2) on residuals       0.0240
24 crudeheat_lag1 with ARIMA(3,1,2) on residuals       0.0240
25      heat_lag1 with ARIMA(3,1,3) on residuals       0.0241
26      heat_lag1 with ARIMA(3,1,1) on residuals       0.0242
27      heat_lag1 with ARIMA(3,1,2) on residuals       0.0243
```

Appendix E: R Code and Reproducibility

All analysis and results in this report were generated using R. The code is organized into separate files corresponding to each stage of the project to ensure clarity and reproducibility.

The following files are included in the submission:

- `stat443-project-EDA.Rmd` — Exploratory data analysis, data cleaning, and visualization
- `stat443-project-persistence-iid.Rmd` — Baseline models (persistence and IID mean)
- `stat443-project-expsmo.Rmd` — Exponential smoothing models
- `stat443-project-arima.Rmd` — ARIMA model development and evaluation
- `stat443-project-arimax.Rmd` — ARIMAX modeling with external variables

The full project code and version history are available on GitHub:

GitHub Repository: Time-Series-Forecasting-of-U.S.-Fuel-Price-Spreads

All code files are fully reproducible and include comments explaining key steps in the analysis.